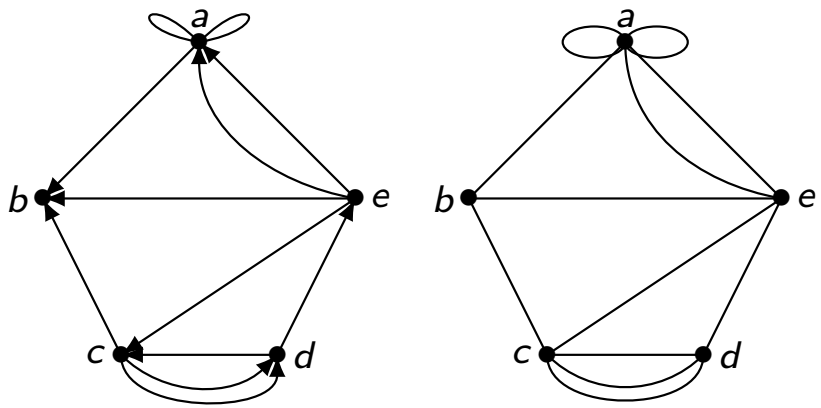
Digraphs I

- A *digraph* $D = (V, A)$ has a set V of elements called *vertices* and a set A of *ordered* pairs of not necessarily distinct vertices called *arcs*. Each arc is of the form

$$\alpha = (a, b) \tag{12.1}$$

  where a and b are vertices. We think of the arc α as *leaving* a and *entering* b, that is, pointed (or directed) from a to b.
- The arc α in (12.1) has *initial vertex* $\iota(\alpha) = a$ and *terminal vertex* $\tau(\alpha) = b$. We may generalize a digraph to a *general digraph* in which multiple arcs are allowed.



Digraphs II

- A vertex x of a general digraph $D = (V, A)$ has two degrees. The *outdegree* of x is the number of arcs of which x is the initial vertex:

$$|\{\alpha|\iota(\alpha) = x\}|.$$

The *indegree* of x is the number of arcs of which x is the terminal vertex

$$|\{\alpha|\tau(\alpha) = x\}|.$$

Theorem

In a general digraph the sum of the indegrees of the vertices equals the sum of the outdegrees, and each is equal to the number of arcs.



Digraphs III

- With any general graph $G = (V, E)$, we can obtain a general digraph $D = (V, A)$ by giving each edge $\{a, b\}$ of E an orientation. Such a digraph D is called an *orientation* of G. A general graph has many different orientations. Conversely, given a general digraph $D = (V, A)$ we can remove the directions of its arcs, thereby obtaining a general graph $G = (V, E)$. Such a graph is called the *underlying general graph* of D.
- An orientation of a complete graph K_n with n vertices is called a *tournament*. The nicest kinds of tournaments are those in which it is possible to order the players in a list

$$p_1, p_2, \cdots, p_n$$

so that each player beats all those further down on the list. Such tournaments are called *transitive tournaments*.



Digraphs IV

- Let $D = (V, A)$ be a general digraph. A sequence of m arcs of the form
$$(x_0, x_1), (x_1, x_2), \cdots, (x_{m-1}, x_m) \tag{12.2}$$
is called a *directed walk of length m from vertex x_0 to vertex x_m*. The *initial vertex* of the walk is x_0 and the *terminal vertex* is x_m. The directed walk is *closed* if $x_0 = x_m$ and *open* otherwise. We also denote the walk (12.2) by
$$x_0 \to x_1 \to x_2 \to \cdots \to x_m.$$
- A directed walk with distinct arcs is a *directed trail*; a directed trail with distinct vertices (except possibly the initial and terminal vertices) is a *path*; a closed path is a *directed cycle*.
- A general digraph is *connected*, provided that its underlying general graph is connected. A general digraph is *strongly connected*, provided that, for each pair of distinct vertices a and b, there is a directed walk from a to b and a directed walk from b to a.



Digraphs V

- A directed trail in a general digraph D is called *Eulerian*, provided that it contains very arc of D. A *Hamilton path* is a path that contains every vertex. A *directed Hamilton cycle* is a directed cycle that contains every vertex.

Theorem

Let D be a connected digraph. Then D has a closed Eulerian directed trail if and only if the indegree of each vertex equals the outdegree.

Theorem

Let D be a connected digraph and let x and y be distinct vertices of D. Then there is a directed directed Eulerian trail from x to y if and only if

1. the outdegree of x exceeds its indegree by 1;
2. the indegree of y exceeds its outdegree by 1;
3. for each vertex $z \neq x, y$, the indegree of z equals its outdegree.



Digraphs VI

Theorem

Let D be a strongly connected digraph without loops. If, for each vertex x, we have

$$(\text{outdegree of } x) + (\text{indegree of } x) \geq n,$$

then D has a directed Hamilton cycle.

Theorem

Every tournament has a Hamilton path.

Theorem

Let $G = (V, E)$ be a connected graph. Then G has a strongly connected orientation if and only if G does not have any bridges.



Digraphs VII

Algorithm for a strongly connected orientation of a bridgeless connected graph $G = (V, E)$

1. Put $U = \emptyset$.
2. Locate a cycle γ of G.
   1. Orient the edges of γ so that it becomes a directed cycle.
   2. Add the vertices of γ to U.
3. While $U \neq V$, do the following:
   1. Locate an edge $\alpha = \{x, y\}$ joining a vertex x in U to a vertex y not in U.
   2. Locate a cycle γ containing the edge α.
   3. Orient the edge α from x to y and continue to orient the edges of γ as if to form a directed cycle until arriving at a vertex z in U.
   4. Add to U all the vertices of γ from x to z.
4. Orient in either direction every edge that has not yet received an orientation.



Digraphs VIII

Example

A trading problem – There are n traders $t_1, t_2, \ldots, t_n$ who enter a market, each with an indivisible item to offer in trade. We assume for simplicity that a trader never has any use for more than one of the items, but except for this assumption, the items are freely transferable from one trader to another. Each trader ranks the n items brought to the market (including his own) according to his preference for them. There are no ties, and thus each trader ranks the items from 1 to n. The effect of the market activity is to redistribute (or permute) the ownership of the items among the n traders. Such a permutation is called an *allocation*. We regard an allocation as a one-to-one function

$$\rho : \{t_1, t_2, \cdots, t_n\} \to \{t_1, t_2, \cdots, t_n\},$$

where $\rho(t_i) = t_j$ means that trader t_i receives the item of trader t_j in the allocation.

An allocation ρ is called a *core allocation*, provided that it has the following property: There does not exist a subset S of fewer than n traders such that,



Digraphs IX

by trading amongst themselves, each receives an item that he ranks more highly than in the allocation ρ. For example, suppose that $n = 5$ and the preferences of the traders are as given by

	t_1	t_2	t_3	t_4	t_5
t_1	4	3	1	2	5
t_2	4	3	1	2	5
t_3	4	3	5	1	2
t_4	1	4	3	5	2
t_5	4	5	2	1	3

One possible allocation ρ is

$$\rho(t_1) = t_2, \rho(t_2) = t_3, \rho(t_3) = t_1, \rho(t_4) = t_5, \rho(t_5) = t_4.$$

This allocation is not a core allocation since $\rho'(t_3) = \rho(t_4) = t_5$, $\rho'(t_4) = \rho(t_3) = t_1$. A core allocation in this case is

$$\rho(t_1) = t_3, \rho(t_2) = t_2, \rho(t_3) = t_4, \rho(t_4) = t_1, \rho(t_5) = t_5.$$



Digraphs X

- We consider a digraph $D = (V, A)$ in which the vertices are the n traders. We put an arc from each vertex to every other, including the vertex itself. Each vertex has indegree equal to n and outdegree equal to n. The digraph D is a *complete digraph* or order n. An allocation corresponds to a partition of the vertices into directed cycles.

Lemma

Let D be a digraph in which each vertex has outdegree at least 1. Then there is a directed cycle in D.



Digraphs XI

Algorithm for a directed cycle

Let u be any vertex.

(1) Put $i = 1$ and $x_1 = u$.

(2) If x_i is the same as one of the previously chosen vertices x_j, $(j < i)$, then go to (4). Else, go to (3).

(3) Do the following.

1. Choose an arc (x_i, x_{i+1}) leaving vertex x_i.
2. Increase i by 1.
3. Go to (2).

(4) Output the directed cycle

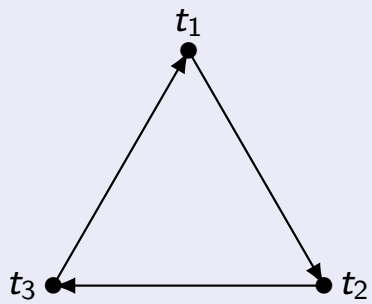
$$x_j \to x_{j+1} \to \cdots \to x_{i-1} \to x_i = x_j.$$



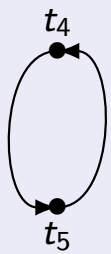
Digraphs XII

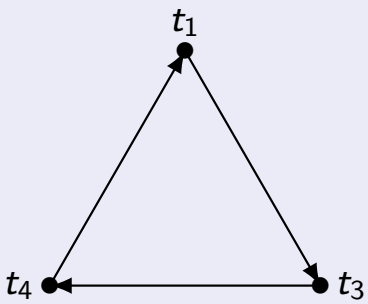
Example

D_ρ and D_{ρ^*} correspond to the allocations ρ and ρ^*. D^1 is the preference digraph,

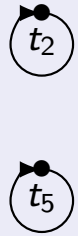


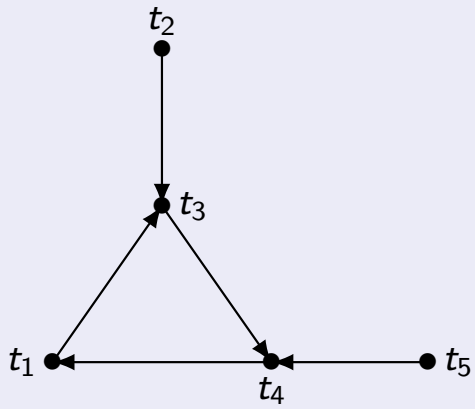
(a) D_ρ





(b) D_{ρ^*}





(c) D^1

Digraphs XIII

Corollary

Let X be a set of n elements and let $f : X \to X$ be a one-to-one function. Let $D_f = (X, A_f)$ be the digraph whose set of arcs is

$$A_f = \{(x, f(x)) : x \in X\}.$$

Then the arcs of D_f can be partitioned into directed cycles with each vertex belonging to exactly one directed cycle.

Theorem

Every trading problem has a core allocation.



Networks I

- A *network* is a digraph (V, A) in which two vertices–the *source* s and the *target* t–are distinguished, where $s \neq t$, and in which each arc α has a nonnegative weight $c(\alpha)$, called its *capacity*. We denote a network by $N = (V, A, s, t, c)$.
- Formally, a *flow* in the network N is defined to be a function f that assigns a real number $f(\alpha)$ to each arc α, subject to the following constraints:
  1. $0 \leq f(\alpha) \leq c(\alpha)$. (The flow through an arc is nonnegative and does not exceed its capacity.)
  2. $\sum_{\iota(\alpha)=x} f(\alpha) = \sum_{\tau(\alpha)=x} f(\alpha)$ for each vertex $x \neq s, t$. (For each vertex x other than the source and the target, the flow into x equals the flow out of x.)

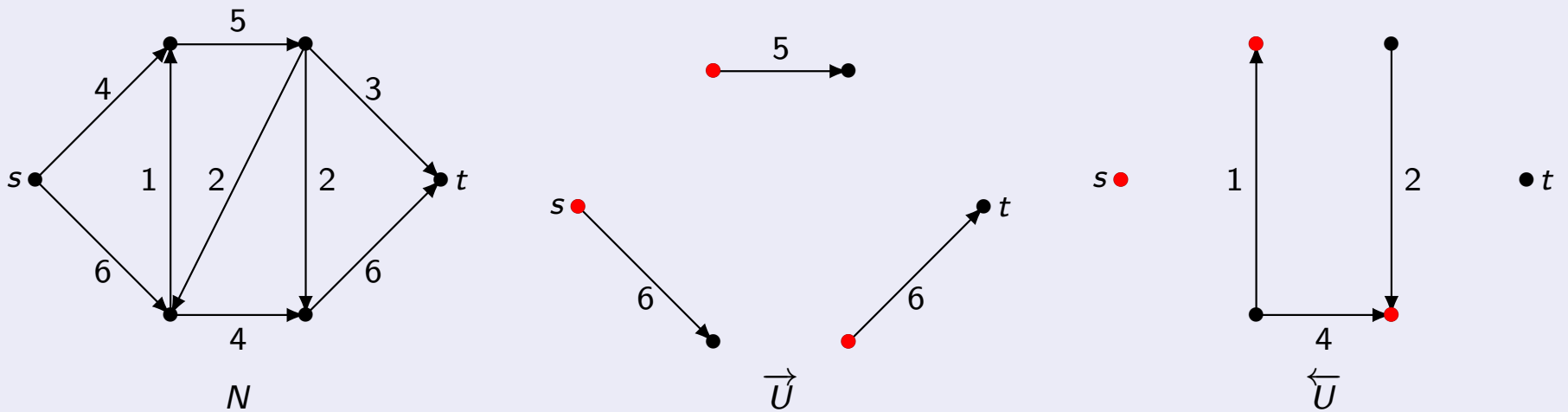


Networks II

- For each set of vertices U, we let

$$\overrightarrow{U} = \{\alpha : \iota(\alpha) \text{ is in } U, \tau(\alpha) \text{ is not in } U\},$$
$$\overleftarrow{U} = \{\alpha : \iota(\alpha) \text{ is not in } U, \tau(\alpha) \text{ is in } U\}.$$

Example



Networks III

Lemma

Let f be a flow in the network $N = (V, A, s, t, c)$ and let U be a set of vertices containing the source s but not the target t. Then

$$\sum_{\alpha \in \overrightarrow{U}} f(\alpha) - \sum_{\alpha \in \overleftarrow{U}} f(\alpha) = \sum_{\iota(\alpha)=s} f(\alpha) - \sum_{\tau(\alpha)=s} f(\alpha).$$

- In the lemma, take $U = V - \{t\}$. Then $\overrightarrow{U}$ is the set of all arcs whose terminal vertex is t, and $\overleftarrow{U}$ is the set of all arcs whose initial vertex is t. Hence,

$$\sum_{\iota(\alpha)=s} f(\alpha) - \sum_{\tau(\alpha)=s} f(\alpha) = \sum_{\tau(\alpha)=t} f(\alpha) - \sum_{\iota(\alpha)=t} f(\alpha). \tag{12.3}$$

  The common value of the two expressions in (12.3) is called the *value of the flow* f and is denoted by $\mathrm{val}(f)$.
- Given a network $N = (V, A, s, t, c)$, a flow in N is a *maximum flow*, provided that it has the largest value among all flows in N.



Networks IV

- A *cut* is a set C of arcs such that each path from the source s to the target t contains at least one arc in C. The *capacity* $\mathrm{cap}(C)$ of a cut C is the sum of the capacities of the arcs in C. A cut is a *minimum cut*, provided that it has the smallest capacity among all cuts in N.
- A cut is a *minimal cut*, provided that each set obtained from C by the deletion of one of its arcs is not a cut.

Lemma

Let $N = (V, A, s, t, c)$ be a network with C a minimal cut. Let U be the set of all vertices x for which there exists a path from the source s to x that contains no arc in C. Then $\overrightarrow{U}$ is a cut and $C = \overrightarrow{U}$.



Networks V

Theorem

Let $N = (V, A, s, t, c)$ be a network. Then the maximum value of a flow in N equals the minimum capacity of a cut in N. In other words, the value of a maximum flow equals the capacity of a minimum cut. If the capacities of all the arcs are integers, then there is a maximum flow all of whose values are integers as well.

Basic Flow Algorithm

Begin with any integer-valued flow f (The *zero flow*, in which all flow values equal zero, will suffice), on the network $N = (V, A, s, t, c)$.

(0) Set $U = \{s\}$.

(1) While there exists an arc $\alpha = (x, y)$ with either

- (a) x in U, y not in U, and $f(\alpha) < c(\alpha)$, or
- (b) x not in U, y in U, and $f(\alpha) > 0$, put y in U (in case of (a)) or put x in U (in case of (b)).

(2) Output U.

Networks VI

- The algorithm results in one of the following two possibilities.

  Breakthrough: An integer-valued flow f' has been found with $\mathrm{val}(f') = \mathrm{val}(f) + 1$. In this case, we repeat the algorithm with $f = f'$.
  Nonbreakthrough: Breakthrough has not occurred. In this case, we exhibit a cut whose capacity equals the value of the flow f. The cut is our desired minimum cut $\hat{C}$ and the flow f is our desired maximum flow $\hat{f}$.

- Case 1: The target t is in U.

$$x_0 = s, x_1, x_2, \ldots, x_{m-1}, x_m = t$$

such that, either

(a) $\alpha_j = (x_j, x_{j+1})$ is an arc of the network with $f(\alpha_j) < c(\alpha_j)$ or
(b) $\alpha_j = (x_{j+1}, x_j)$ is an arc of the network with $f(\alpha_j) > 0$.

We now define an integer-valued function f' on the set A of arcs by

$$f'(\alpha) = \begin{cases} f(\alpha) + 1 & \text{if } \alpha \text{ is one of the arcs } \alpha_j \text{ in (a)}, \\ f(\alpha) - 1 & \text{if } \alpha \text{ is one of the arcs } \alpha_j \text{ in (b)}, \\ 0 & \text{otherwise.} \end{cases}$$



Networks VII

- Case 2: The target t is not in U. $\overrightarrow{U}$ is a cut in this case and it follows that

$$\mathrm{val}(f) = \mathrm{cap}\,\overrightarrow{U}.$$

Hence, $\hat{f} = f$ is a maximum flow and $\hat{C} = \overrightarrow{U}$ is a minimum cut.

Theorem

Let s and t be distinct vertices of a digraph $D = (V, A)$. Then the maximum number of pairwise arc-disjoint paths from s to t equals the minimum number of arcs in an st-separating set.

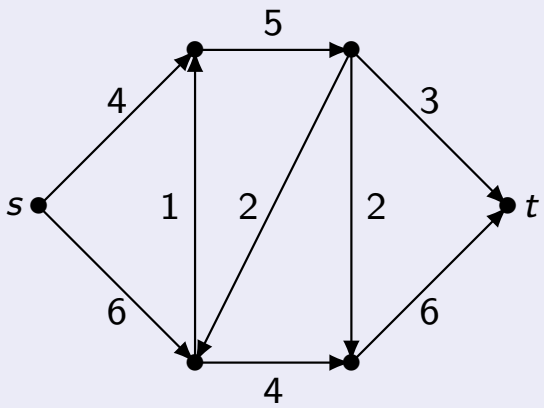
Theorem

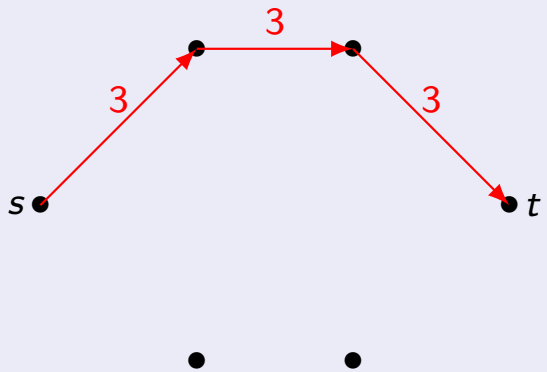
Let G be a bipartite graph. Then $\rho(G) = c(G)$.



Networks VIII

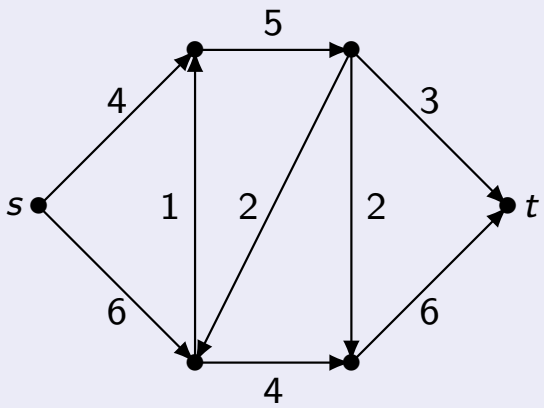
Example

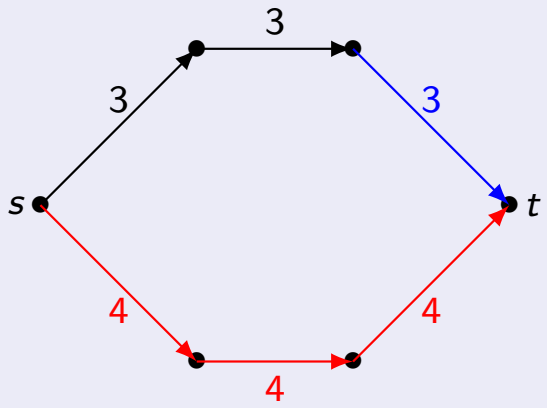




Networks VIII

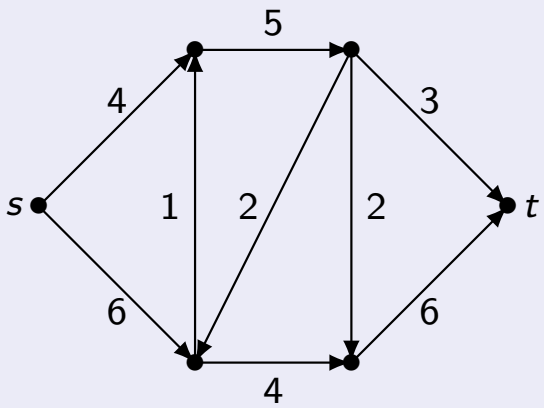
Example

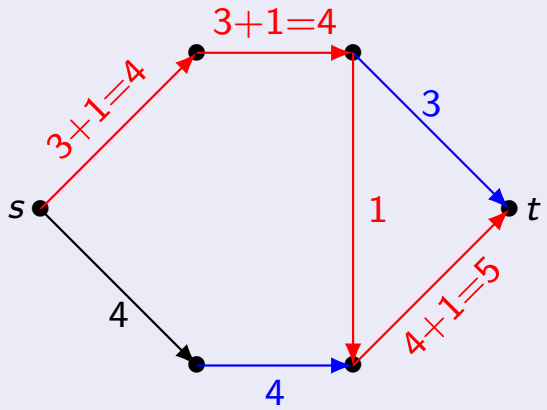




Networks VIII

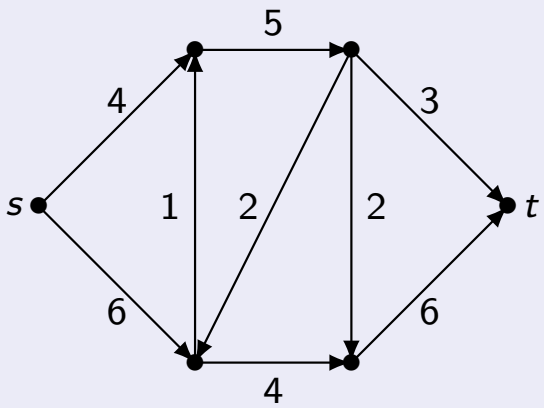
Example

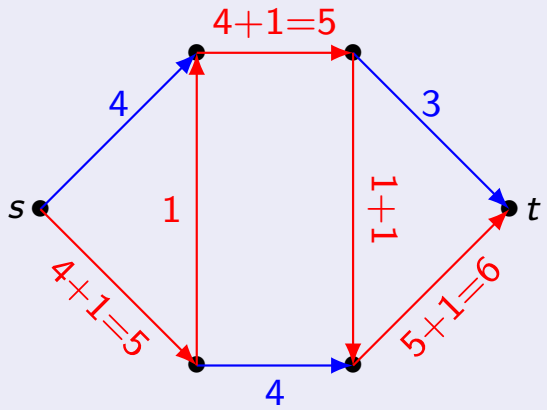




Networks VIII

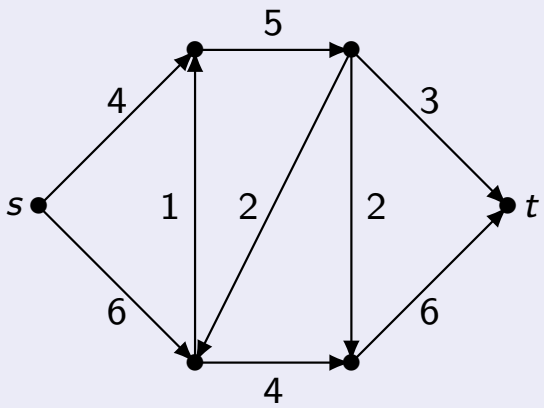
Example

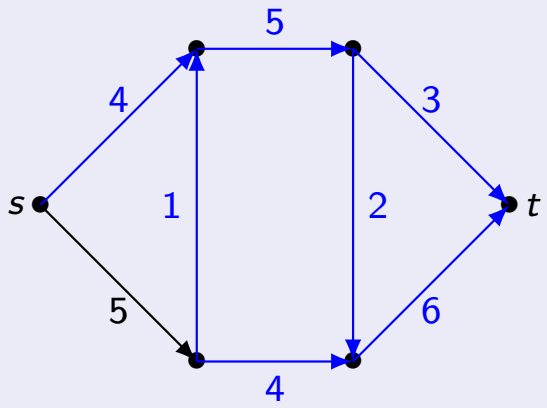




Networks VIII

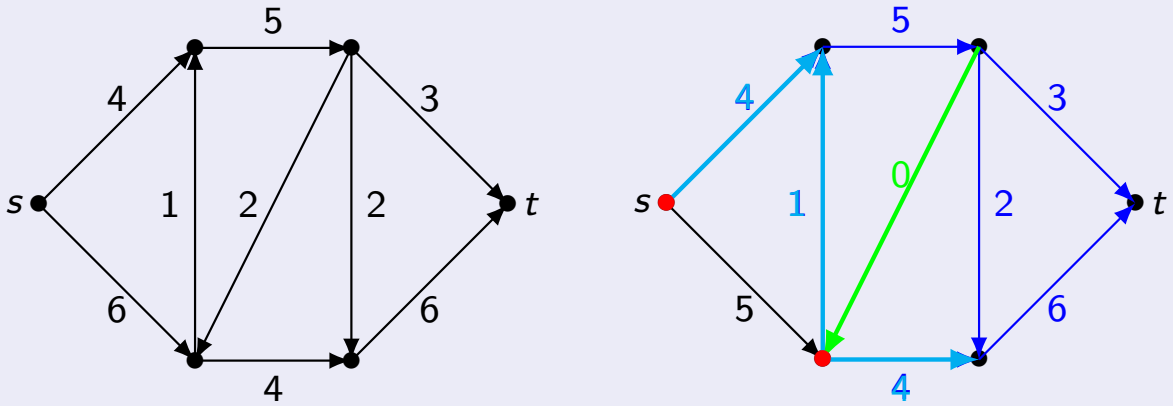
Example





Networks VIII

Example



$$\mathrm{val}(f) = \sum_{\alpha \in \overrightarrow{U}} f(\alpha) - \sum_{\alpha \in \overleftarrow{U}} f(\alpha) \leq \sum_{\alpha \in \overrightarrow{U}} c(\alpha) = \mathrm{cap}\,\overrightarrow{U}.$$